

BLUESTOCK MUTUAL FUND ANALYTICS FINAL PERFORMANCE & ANALYTICS REPORT

An End-to-End Analytics Lifecycle: From Data Engineering and ETL Pipelines to Machine Learning Risk Ratios & Executive Business Intelligence

Prepared by: Aditya Verma

intern Data analytics

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1. Executive Summary

The Indian mutual fund industry has undergone an unprecedented transformation over the past decade, driven by the structural financialization of household savings, widespread digital onboarding, and a monumental shift in retail sentiment. Historically anchored to physical assets like real estate and gold, individual wealth is rapidly moving into capital markets. The primary mechanism driving this democratization is the Systematic Investment Plan (SIP), which provides regular, disciplined equity market access to millions of small-scale investors across the country.

The **Bluestock Mutual Fund Analytics** project was established to engineer, validate, and present an end-to-end analytical stack capable of bridging the gap between raw, disconnected market disclosures and institutional-grade business intelligence. This comprehensive report consolidates the data models, repeatable ETL pipelines, core exploratory insights, advanced risk-adjusted return rankings, and corporate dashboard outputs into a single, cohesive document.

1.1 Core Objectives & Analytical Framework

The platform is engineered to resolve five core business and technical challenges:

- **Pipeline Reliability and Repeatability:** Can a mixed dataset of daily NAV histories, benchmark levels, and investor transactions be programmatically cleaned and integrated into an optimized SQLite relational database without referential errors?
- **Market Participation Dynamics:** What do macroeconomic industry markers—such as monthly SIP velocities and category net flows—indicate about overall systemic market liquidity and investor stability?
- **Granular Behavioral Demographics:** How do geographic location (T30 vs. B30 city tiers), age groups, income levels, and digital payment frameworks impact mutual fund transaction frequency and size?
- **Risk-Adjusted Suitability:** Which mutual fund schemes demonstrate genuine active management capabilities (Alpha) and superior risk-mitigation efficiency once absolute returns are adjusted for standard deviation, downside risk, and peak-to-trough drawdowns?
- **Interactive Business Intelligence:** How can highly technical database relations be translated into an intuitive, multi-page executive dashboard that facilitates data exploration without misleading users or overstating precision?

1.2 Headline Empirical Findings

The foundational data architecture, exploratory findings, and quantitative models have revealed several vital, interconnected truths about the current analytical sample:

- **The Evaluated Universe:** The project controls a curated sample of 40 mutual fund schemes representing 10 leading asset management companies (AMCs). This includes 34 equity-oriented

and 6 debt/liquid schemes. Structurally, regular plans comprise 80% (32 schemes) while direct plans account for 20% (8 schemes), forming an effective base for cost-impact analysis.

- **Explosive SIP Inflow Inflows:** Month-on-month industry SIP inflows demonstrate strong, uninterrupted growth, moving from INR 13,856 crore in January 2023 to INR 31,002 crore in December 2025—a stellar 123.7% expansion. Concurrently, the accumulation of market gains and regular deposits pushed total SIP Assets Under Management (AUM) up by 173.2% over the same timeline.
- **The Retail Participation vs. Volume Divergence:** SIP allocations represent 60.2% of all transaction records but contribute only 6.2% of the aggregate dollar transaction value. This proves empirically that SIPs serve as the primary retail vehicle for market entry and long-term participation, while short-term institutional asset movement is dominated by high-ticket lumpsum buys and redemptions.
- **The High-Return, High-Volatility Trade-Off:** Small-cap equity schemes dominate absolute short-term performance charts, recording a one-year average return of 22.26% and a three-year average return of 21.69%. However, this outperformance is linked to a severe 25% average annualized volatility and historical drawdowns exceeding -28%. Conversely, liquid assets generate lower absolute returns but deliver near-zero volatility and superior downside risk protection.
- **The Integrated Leaderboard:** The platform's multi-factor composite scorecard ranks the Mirae Asset Large Cap Fund (Regular) first, closely followed by the Kotak Flexicap Fund (Regular) and the ICICI Prudential Midcap Fund (Regular). This model successfully balances five dimensions: multi-year compound growth, total risk-adjusted efficiency (Sharpe), downside optimization (Sortino), alpha generation, and low operational fees.

1.3 Strategic Implementation Roadmap

To scale this analytics prototype into a resilient, corporate-grade system, development should proceed along three structured paths:

- **Automated Ingestion & Merging:** Replace the static CSV rebuild process with an automated, script-driven delta fetcher. This architecture will extract daily NAV updates from public APIs and merge them seamlessly into the core historical tables while removing duplicate compound records.
- **Benchmark Matching Enrichment:** Upgrade the performance math to calculate volatility, beta, and alpha against individual, fund-specific benchmarks (e.g., Nifty Midcap 150 for mid-cap schemes) rather than a single market index. This prevents tracking errors from distorting active manager alphas.
- **Enterprise Infrastructure Controls:** Migrate the local SQLite database to an institutional multi-user PostgreSQL engine. This pipeline should be scheduled using Apache Airflow to run automated quality assurance checks before triggering an automatic data refresh on the Power BI server.

2. Data Architecture and Quality Audits

The analytics stack is fed by a structured data layer containing ten source CSV datasets, located within the processed data repository. These files encompass fund identities, micro-level performance tracking, macro industry flows, and detailed client transaction logs.

2.1 Source Dataset Inventory

Dataset Name	Rows	Cols	Primary Analytical Application
dim_fund (Fund Master)	40	15	Establishes identity, asset class tier, plan type, declared benchmark index, and expense ratios.
fact_nav (NAV History)	46,000	3	Stores a deep time-series of daily Net Asset Values for trend and return calculation.
fact_performance (Scheme Performance)	40	19	Houses historical point-in-time metrics including Sharpe, Sortino, alpha, beta, and drawdowns.
fact_transactions (Investor Transactions)	32,778	13	Logs transaction types, order sizing, geographical distributions, demographics, and KYC status.
fact_portfolio_holdings	322	8	Details individual equity holdings, underlying sector weights, and stock valuations.
fact_benchmark_returns	8,050	3	Tracks the daily closing levels of essential indices including NIFTY50 and NIFTY100.
fact_sip_inflows	36	6	Aggregates monthly industry active accounts, total recurring inflows, and global SIP AUM.
fact_category_inflows	144	3	Monitors systemic monthly capital flows across Equity, Debt, and Thematic styles.
fact_aum	90	5	Tracks historical asset movements and product counts at the asset management house level.
fact_industry_folios	21	6	Measures net monthly folio additions grouped by underlying asset categorization.

2.2 Spatial-Temporal Coverage and Integrity Keys

The database provides broad historical coverage. The daily asset NAV records span from **January 3, 2022, through May 29, 2026**, accumulating exactly 1,150 individual trading day observations for each of the 40 mutual fund schemes. This timeline captures a significant period of market volatility, encompassing major bull market extensions and sharp corrections. In parallel, the investor transaction log captures micro-activity over a 17-month window stretching from January 1, 2024, through May 30, 2025.

Relational integrity is maintained by using the Association of Mutual Funds in India (AMFI) code as the primary, unique logical key across the schema. The `amfi_code` connects the fund master dimension to daily asset prices, stock allocations, performance tracking, and transaction records. All calendar indicators are standardized as **YYYY-MM-DD** strings to optimize date-filtering queries within SQLite.

2.3 Data Quality Controls and Verification Audits

Before loading the relational schema, all processed data underwent automated validation tests to enforce data quality and consistency:

- **Referential Key Coverage (Pass):** 100% of the AMFI identifiers inside the daily NAV records, performance tracking tables, stock holdings, and client transaction logs match a valid core identity in the master table.
- **NAV Boundaries & Completeness (Pass):** Zero null, missing, or negative values were found across the daily price histories. All daily entries contain positive, non-zero values.
- **Key Uniqueness Verification (Pass):** A compound unique constraint applied to the combination of AMFI code and date confirmed that no duplicate entries exist in the daily pricing table.
- **Transaction Sizing Constraints (Pass):** The transaction ledger contains no negative amounts. Outflows are tracked as positive absolute values with a type indicator set to 'Redemption'.
- **Portfolio Allocation Boundaries (Pass):** Aggregated security weights for each scheme sum closely to 100%. Small residual deviations reflect cash cushions or short-term liquid reserves.

CRITICAL DESIGN LIMITATION & REPRESENTATIVE BIAS

The 40-scheme analytical universe is carefully selected for portfolio development and does not represent a full census of the Indian mutual fund market. Equity funds account for 85% of the entries, and regular intermediary plans comprise 80% of the sample. Accordingly, category-level insights in this report reflect the project's specific dataset and should not be used as exhaustive estimates for the broader mutual fund industry.

3. ETL Pipeline Architecture and Operational Controls

The data platform features a modular, script-driven ETL architecture coordinated by a master orchestration script (`run_pipeline.py`). This layout separates core extraction, notebook cleaning, risk-engine math, schema modeling, and integration verification into isolated components.

3.1 Sequential Processing Stages

Stage	Core Component	Operational Purpose & Actions
1	Source Ingestion	Loads the ten historical CSV datasets and extracts optional live NAV updates via API endpoints.
2	Data Cleaning	Standardizes date records, confirms unique identifiers, replaces null values, and handles numeric data types.
3	Metric Calculation	Computes annualized historical returns, daily standard deviations, risk metrics, and drawdown depths.
4	Schema Compilation	Executes strict table definitions, applies value constraints, and builds foreign keys inside SQLite.
5	Database Loading	Appends processed and derived datasets into 11 distinct relational tables using bulk commits.
6	Relational Validation	Runs SQLite internal checks, foreign-key audits, row counts, and ten verification queries.

3.2 Live NAV Extraction Infrastructure

The ingestion engine includes an operational script (`live_nav_fetch.py`) designed to query active price endpoints at `mfapi.in`. To protect public API stability, the fetcher enforces a polite **0.5-second connection delay** and utilizes strict request timeouts. To ensure stability in isolated environments, live network extraction is optional; the default orchestrator relies on processed local data unless explicitly triggered via the `--fetch-live-nav` parameter.

3.3 Database Relational Schema and Performance Indexes

The generated database (`bluestock_mf_pipeline.db`) uses strict physical schema constraints to protect data integrity. Tables utilize explicit primary keys and enforce foreign key checks. To ensure

query speed across the 46,000 daily pricing observations and 32,778 customer transactions, the database creates dedicated relational indexes during initialization:

- `idx_nav_date` built on **fact_nav(date)** – Optimizes performance for monthly rolling returns and cross-scheme average calculations.
- `idx_transactions_date` built on **fact_transactions(transaction_date)** – Speeds up demographic trend sorting and chronological analysis.
- `idx_transactions_state` built on **fact_transactions(state)** – Accelerates geographical capital flow aggregations.
- `idx_holdings_sector` built on **fact_portfolio_holdings(sector)** – Optimizes industry concentration and sector risk exposure dashboards.

3.4 Fail-Fast Mechanics and Deployment Logs

The execution pipeline includes strict operational checks to ensure stability. If a source file is missing, the orchestrator triggers an immediate exception and terminates execution to prevent data corruption. Furthermore, database schema constraints block any non-positive NAV entries or negative order sizes. As a final integration test, the pipeline executes ten analytical business queries from `sql/queries.sql`; any query failure halts the deployment.

The deployment logs verify successful database integration and pipeline completion:

```
INFO: Stage 1/5: live NAV download skipped
INFO: Stage 2/5: calculating derived fund metrics
INFO: Wrote metrics for 40 funds to data/processed/derived_fund_metrics.csv
INFO: Stage 3/5: rebuilding the SQLite schema
INFO: Database schema ready at data/db/bluestock_mf_pipeline.db
INFO: Stage 4/5: loading processed datasets
INFO: Loaded 40 rows into dim_fund
INFO: Loaded 46000 rows into fact_nav
INFO: Loaded 32778 rows into fact_transactions
INFO: Stage 5/5: validating the database
INFO: Validated 11 tables containing 87561 rows
INFO: Executed and verified 10 analytical integration queries
INFO: Pipeline completed successfully
```

4. Exploratory Data Analysis (EDA) Findings

Exploratory analysis of the relational layer reveals critical structural trends, asset concentrations, and retail investor behaviors across the market sample.

4.1 Asset Concentration Among Top Fund Houses

An analysis of the asset distribution reveals high concentration among the leading fund houses. The five largest asset management firms control a dominant share of total assets under management within the project dataset:

- **SBI Mutual Fund:** Leads the asset snapshot with a massive volume of INR 12.50 lakh crore (31.1% market share within the dataset).
- **ICICI Prudential Mutual Fund:** Ranks second, managing a substantial asset base of INR 10.74 lakh crore (26.7% market share).
- **HDFC Mutual Fund:** Positions closely behind, controlling INR 9.30 lakh crore (23.1% market share).
- **Nippon India Mutual Fund:** Maintains an asset volume of INR 4.01 lakh crore (10.0% market share).
- **Kotak Mahindra Mutual Fund:** Holds an asset volume of INR 3.68 lakh crore (9.1% market share).

Together, the top three asset managers direct a concentrated **80.9% of total assets** represented in the sample. This concentration highlights significant distribution and scale advantages for the leading AMCs. For retail investors, a small group of fund managers directs the majority of capital, which can impact liquidity in underlying holdings.

4.2 Systematic Investment Plan (SIP) Dynamics

The industry data records explosive growth in retail participation over the evaluated three-year timeline:

- **Monthly Volume Extensions:** Regular monthly SIP inflows expanded by 123.7%, rising from INR 13,856 crore in January 2023 to INR 31,002 crore by December 2025.
- **Active Accounts Growth:** Total registered, active retail accounts grew by 52.5%, climbing from 4.91 crore to 7.49 crore entries.
- **Compounding Asset Accumulation:** Total cumulative SIP AUM increased by 173.2%, rising from INR 4.80 lakh crore to INR 13.12 lakh crore.

The difference between active account growth (52.5%) and total asset growth (173.2%) reflects the compounding effect of market gains and sustained regular contributions. The latest category-flow snapshot shows particularly strong net inflows into liquid funds, sectoral/thematic equity, and mid-cap categories. This suggests simultaneous demand for short-term liquidity management and growth-oriented equity exposure.

4.3 Micro Demographics and Transaction Ledgers

Analyzing the 32,778 transaction lines (representing INR 3.52 billion in aggregate volume) highlights a sharp divergence between transaction frequency and financial value:

- **The SIP Paradox:** SIP allocations comprise 60.2% of all individual transaction records but account for only 6.2% of total transaction value. This confirms that SIPs are primarily tools for disciplined retail participation, whereas high-ticket asset movement occurs via lumpsum and redemption channels.
- **Geographic Tiers:** Top 30 (T30) cities generate 66.3% of transactions and 65.9% of financial value. Beyond 30 (B30) locations contribute approximately one-third of overall market volume, indicating significant and growing retail adoption in smaller cities.
- **Regional Centers:** Punjab, Tamil Nadu, Madhya Pradesh, Rajasthan, and Gujarat emerge as the top five states by transaction value within the sample dataset.
- **Compliance and Infrastructure:** 78.4% of recorded transactions were executed by KYC-verified accounts. Net banking and automated mandate frameworks handle high-value lumpsum entries, while UPI serves as the dominant channel for low-ticket retail transactions.

5. Comprehensive Fund Performance & Risk Analysis

To establish objective comparisons across the 40-fund universe, the computation engine analyzes performance across returns, volatility, benchmark sensitivity, and fee structures.

5.1 Quantitative Ratios and Formulations

The risk engine evaluates performance using the following mathematical models:

1. Compound Annual Growth Rate (CAGR):
$$\text{CAGR} = (\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(365.25 / \text{Days})} - 1$$
2. Annualized Volatility (Standard Deviation):
$$\sigma_{\text{ann}} = \sqrt{ (1 / (N-1)) * \sum (R_t - R_{\text{mean}})^2 } * \sqrt{252}$$
3. Sharpe Ratio (Risk-Adjusted Efficiency):
$$\text{Sharpe} = (R_p - R_f) / \sigma_{\text{ann}} \quad [\text{where } R_f = 6.5\%]$$
4. Capital Asset Pricing Model (Alpha & Beta Sensitivity):
$$R_t - R_f/252 = \alpha_{\text{daily}} + \beta(R_{m,t} - R_f/252) + \epsilon_t$$
$$\beta = \text{Cov}(R, R_m) / \text{Var}(R_m) \quad | \quad \alpha_{\text{annual}} = \alpha_{\text{daily}} * 252$$
5. Maximum Drawdown (Historical Depth Check):
$$\text{Drawdown}_t = \text{NAV}_t / \text{Max}(\text{NAV}_i) - 1 \quad | \quad \text{MDD} = \text{Min}(\text{Drawdown}_t)$$

5.2 Return Performance & Volatility Analysis

Absolute short-term returns are heavily dominated by small-cap funds, with the category averaging a one-year return of 22.26% and a three-year return of 21.69%. This outperforms the large-cap sector's returns of 13.72% (1Y) and 12.99% (3Y). This return premium is accompanied by high volatility: small-cap schemes average 25% annualized volatility, whereas large-cap schemes average 14% volatility, and mid-cap configurations track at 19%. This relationship highlights the necessity of adjusting returns for risk. An investor selecting a small-cap fund based solely on historical returns must be prepared to accept twice the volatility of a large-cap index fund.

AMFI Code	Scheme Name	Catego ry	1Y Return	3Y CAGR	Volatili ty	Shar pe	Risk Grade
119598	SBI Small Cap Fund - Regular	Equity	24.56%	23.39%	25.0%	0.94	Very High
119599	SBI Small Cap Fund - Direct	Equity	20.59%	23.14%	25.0%	0.93	Very High
101207	ABSL Small Cap Fund -	Equity	24.93%	22.38%	24.7%	0.90	Very

	Regular						High
119095	Axis Small Cap Fund - Regular	Equity	21.97%	20.98%	23.8%	0.84	Very High
118634	Nippon India Small Cap - Regular	Equity	21.30%	20.15%	25.2%	0.81	Very High
149324	DSP Small Cap Fund - Regular	Equity	20.20%	20.08%	25.1%	0.80	Very High
125498	HDFC Mid-Cap Opp - Direct	Equity	19.98%	15.29%	19.1%	0.80	High

5.3 Risk-Adjusted and Downside Protection Metrics

Adjusting for volatility changes the asset leaderboards. Liquid funds generate the highest Sharpe ratios (averaging 6.33) because their standard deviation of daily returns is extremely low (0.5%), despite lower absolute returns. Gilt configurations achieve a Sharpe ratio of 1.52 with an average volatility of 4.0%. In comparison, large-cap funds average a Sharpe of 0.88, and small-cap funds track at 0.87. Furthermore, during market corrections, small-cap schemes experienced maximum drawdowns exceeding -28%, while large-cap schemes were limited to -21%, and debt-oriented gilt funds maintained drawdowns under -2.3%. Only penalizing downside volatility using the Sortino ratio favors stable large-cap vehicles over volatile small-cap assets (e.g., SBI Bluechip Fund achieves a Sortino of 2.14).

5.4 Multi-Factor Scorecard Formulations and Standings

To eliminate single-metric bias, the platform implements a multi-factor composite scorecard that ranks schemes across five dimensions: 3-Year CAGR rank (30%), Sharpe ratio rank (25%), Alpha generation rank (20%), Expense ratio rank (15% in ascending order), and Maximum Drawdown rank (10%). The resulting weighted ranks are normalized on a 0 to 100 scale:

$$\text{Raw Score} = 0.30 \cdot R_{\text{cagr}} + 0.25 \cdot R_{\text{sharpe}} + 0.20 \cdot R_{\text{alpha}} + 0.15 \cdot R_{\text{expense}} + 0.10 \cdot R_{\text{drawdown}}$$

$$\text{Normalized Score} = \left[\frac{(\text{Raw Score}_{\text{max}} - \text{Raw Score})}{(\text{Raw Score}_{\text{max}} - \text{Raw Score}_{\text{min}})} \right] \cdot 100$$

Rank	Mutual Fund Scheme Name	Fund House AMC	Plan	3Y CAGR	Sharpe	Alpha	Score
1	Mirae Asset Large Cap Fund -	Mirae Asset	Regul	34.00	1.44	27.0	100.0

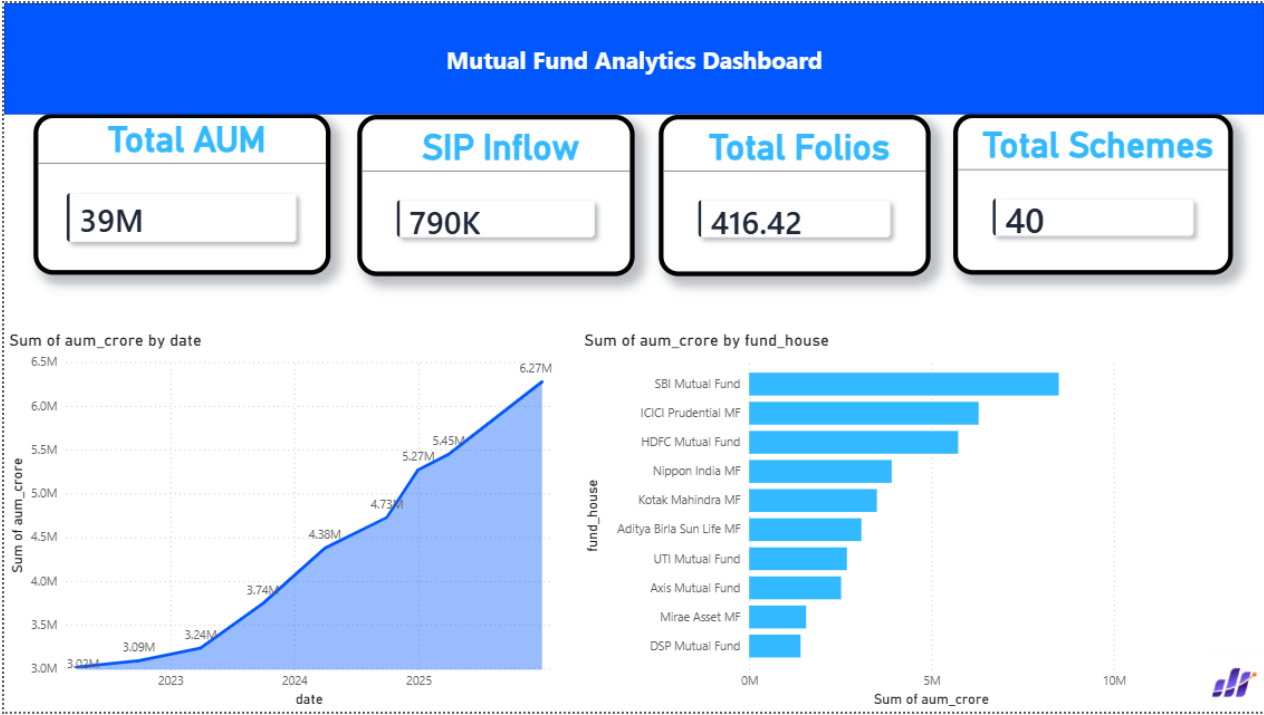
	Regular	MF	ar	%	8	%	
2	Kotak Flexicap Fund - Regular	Kotak Mahindra	Regular	29.58 %	1.307	27.3 %	94.1
3	ICICI Pru Midcap Fund - Regular	ICICI Prudential	Regular	31.78 %	1.180	29.3 %	93.4
4	HDFC Mid-Cap Opportunities - Regular	HDFC Mutual Fund	Regular	32.44 %	1.094	27.2 %	91.8
5	ICICI Pru Bluechip Fund - Direct	ICICI Prudential	Direct	32.49 %	1.027	13.9 %	90.2

6. Executive Dashboard Architecture & Walkthrough

The reporting layer is supported by a comprehensive, four-page Power BI dashboard (bluestock_mf_dashboard.pbix) designed to track asset volumes, performance histories, demographics, and market trends.

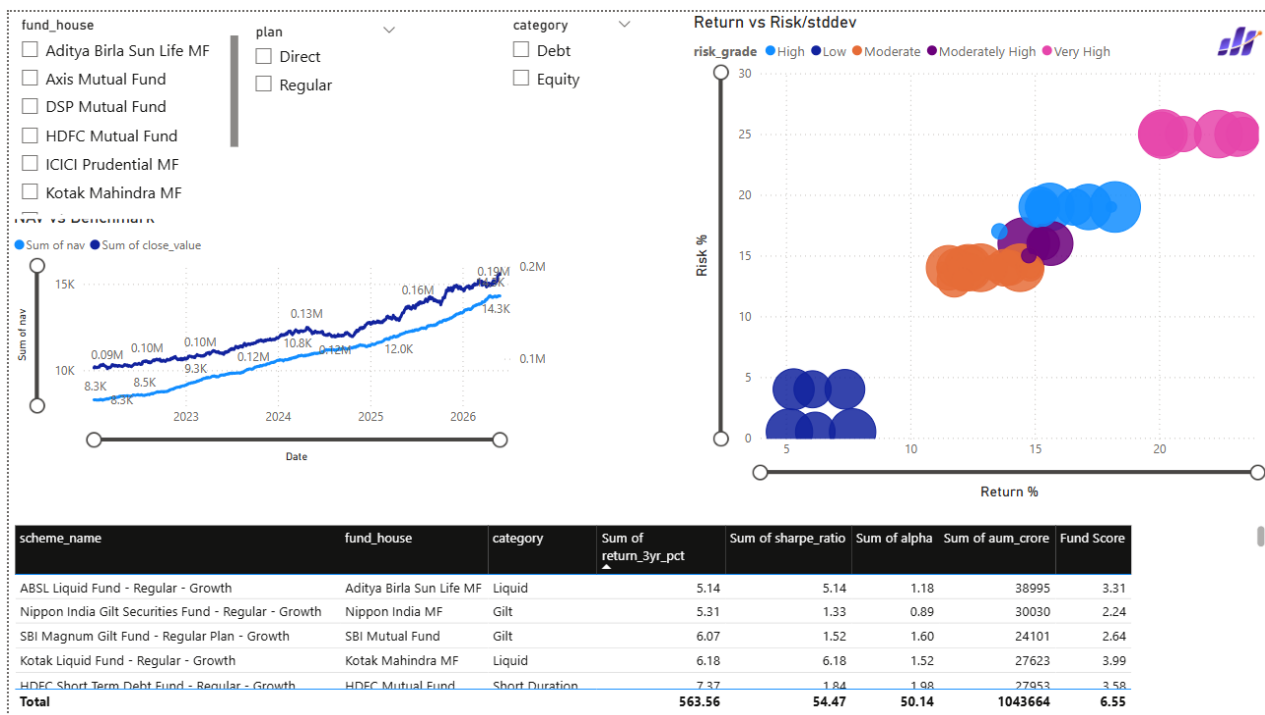
6.1 Page 1: Industry Overview

This interface provides an executive overview of the mutual fund industry, tracking total assets, folio growth, and market shares. Core components include KPI cards detailing aggregate represented AUM (39M), total active accounts (790K), and industry folios (416.42). A donut chart details asset distribution across Equity, Debt, and Hybrid styles, while a horizontal bar chart highlights asset concentration among the leading fund houses. Slicers allow executives to filter the overview by AMC name, scheme category, or plan type.



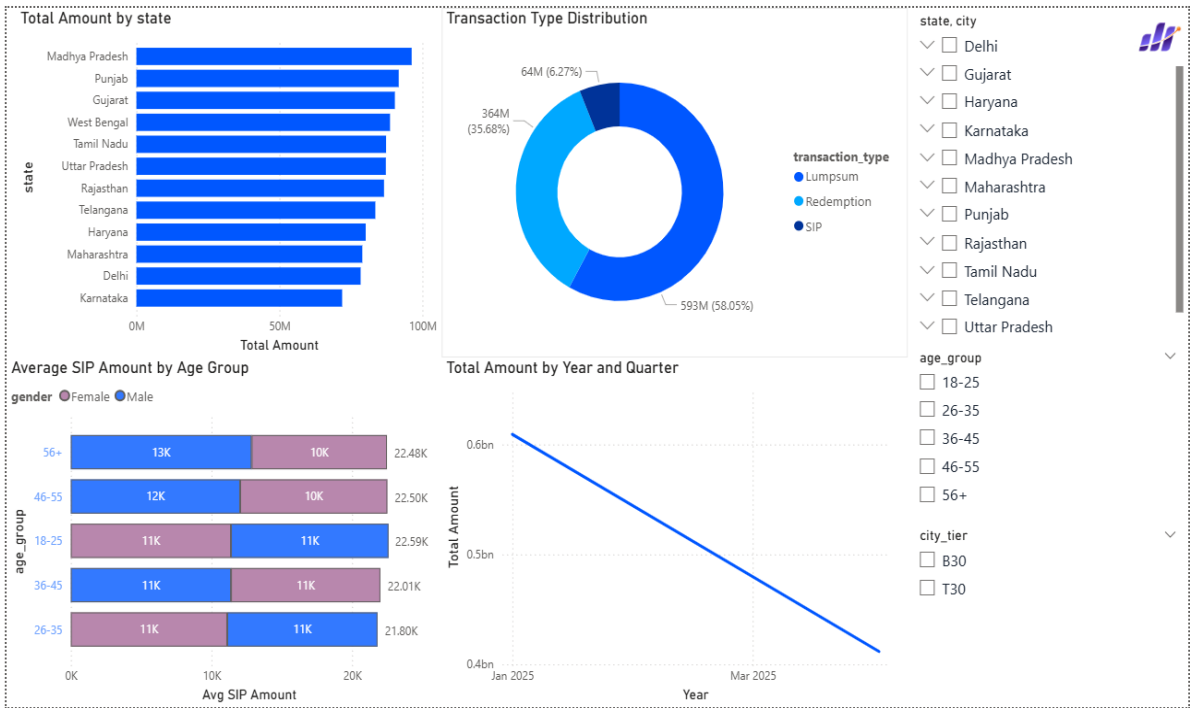
6.2 Page 2: Scheme Performance Analytics

This screen enables detailed comparison of individual mutual fund schemes across returns and risk metrics. A central scatter plot charts annualized returns against daily standard deviations, mapping risk-grade bands to specific visual colors (Low, Moderate, High, Very High) while bubble sizing indicates scheme AUM. An interactive data matrix table lists multi-year returns, alpha generation, beta sensitivity, Sharpe ratios, and composite scorecard results, allowing filters for regular vs. direct plans.



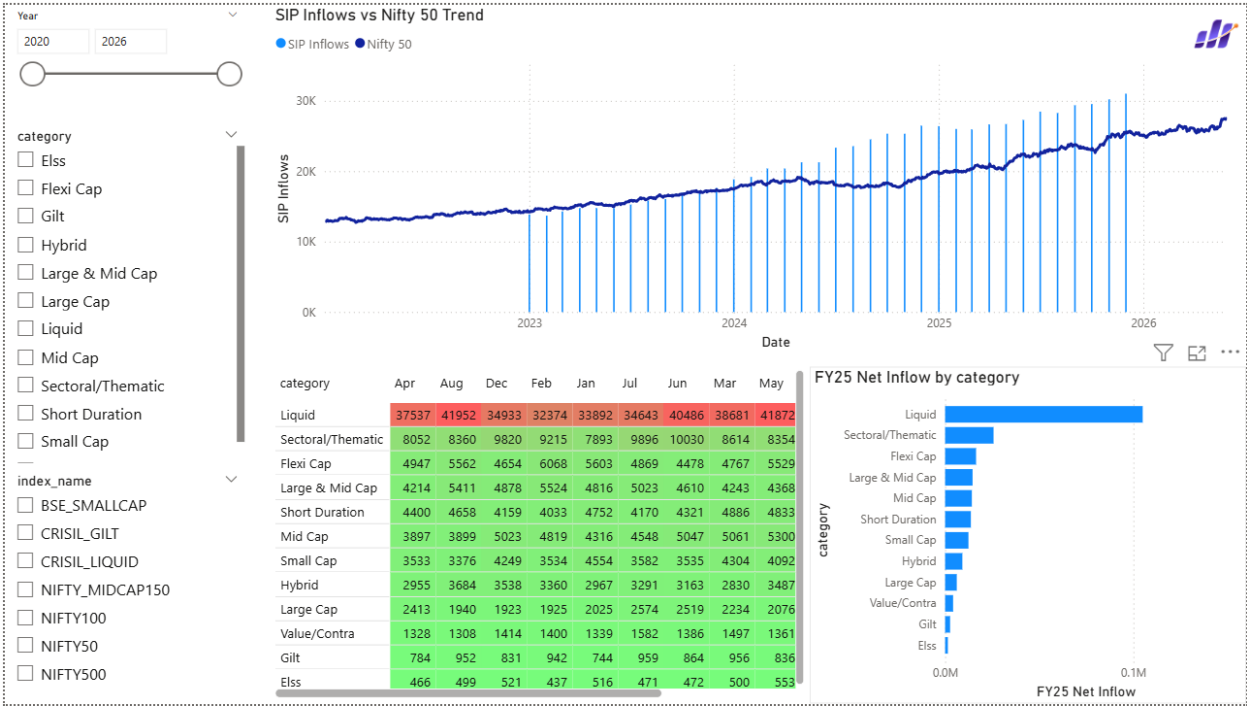
6.3 Page 3: Investor Analytics

This layout analyzes investor transaction volume, geographic distribution, and client demographics. A geographical bar map breaks down transaction value across states, identifying Punjab and Madhya Pradesh as key regions. Dual-segmented pie visuals show a clear contrast: SIP accounts drive 60.2% of transaction counts but comprise only 6.2% of financial value. Slicers allow filtering by KYC status, city tier (T30 vs. B30), and payment mode (UPI, Net Banking, Mandate).



6.4 Page 4: Strategic SIP Trend Monitoring

The final interface tracks monthly SIP trends, account registration rates, and net category inflows. A dual-axis line chart displays the growth of monthly SIP inflows against active retail accounts. An area chart highlights cumulative SIP AUM growth over the three-year period. A net-inflow heat matrix color-codes month-on-month category shifts, highlighting steady capital inflows into liquid and mid-cap schemes.



7. Strategic Recommendations & System Enhancements

Based on an analysis of the codebase, data schema, and performance models, we recommend the following enhancements:

7.1 Data Engineering & Pipeline Infrastructure

- **Automate Live Ingestion Merges:** Deploy a Python delta merge module that automatically combines daily extracts from live APIs into the historical files while removing duplicate entries.
- **Externalize Hardcoded Configurations:** Move parameters such as the 6.5% risk-free rate, timeouts, and local file paths out of script files and into a centralized configuration file (config.yaml).
- **Upgrade to Production Database Backend:** Migrate the local SQLite database to an institutional PostgreSQL engine or cloud data warehouse (e.g., BigQuery) to support concurrent write connections and user access control.

7.2 Analytics & Portfolio Management Recommendations

- **Segment Shortlists by Risk Profile:** Avoid presenting a single top-ranked fund list. Financial advisors should present separate recommendations for low, moderate, high, and very-high-risk profiles.
- **Contextualize Returns with Risk Metrics:** To prevent investors from chasing short-term returns, every chart displaying absolute returns must display corresponding annualized volatility, beta, and maximum drawdown metrics.
- **Implement Fund-Specific Benchmarks:** The current model calculates alpha and beta for all funds against the Nifty 100 Index. We recommend matching each fund's return series to its declared benchmark (e.g., Nifty Midcap 150 for mid-cap funds) to calculate accurate active alphas.

8. Platform Architectural Limitations

This analytics platform provides a solid framework for mutual fund analysis. However, users must be aware of the following structural limitations:

- **Curated Dataset Sample Bias:** The 40-scheme sample is curated for this capstone project and is not a complete representation of the Indian mutual fund industry. Category-level findings represent the project dataset only.
- **Synthetic Transaction Modeling:** The investor transaction dataset is synthetically generated for analytical testing. While it reflects realistic patterns, it should be used for demographic sorting rather than audited market-share claims.
- **Historical Returns Limitation:** All return, Sharpe, and Sortino ratios are calculated using historical data. These metrics measure past performance and do not predict future returns.
- **Single Index Simplification:** Calculating alpha and beta for all schemes against the Nifty 100 Index can introduce tracking error distortions, particularly for small-cap and debt-oriented funds.
- **Scorecard Metric Correlation:** The composite scorecard includes returns, Sharpe ratio, and alpha. Because these metrics are mathematically correlated, the scorecard may double-count positive performance.
- **Static Portfolio Allocation:** Stock allocations reflect a single point-in-time disclosure date (December 31, 2025). This snapshot does not capture subsequent portfolio turnover or changes in sector exposure.

9. Conclusion

The Bluestock Mutual Fund Analytics project successfully demonstrates an end-to-end data analytics lifecycle. Raw CSV datasets are cleaned, standardized, and enriched with advanced risk-adjusted performance metrics. The data is loaded into a validated SQLite database, queried, and visualized in an interactive Power BI dashboard.

The platform's primary value lies in its reusable framework, which allows users to compare market growth, investor demographics, fees, volatility, and downside risk in one place. By implementing benchmark-aware risk calculations, incremental live API updates, and automated data quality checks, this prototype can be scaled into a production-grade research platform.